Hamdan Real Estate ERP — Software Requirements Specification

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1. Introduction

Hamdan Real Estate ERP is a web system that runs Hamdan Builders & Developers' projects end to end: units, sales, collections, construction costs, stock, partners and accounts. This SRS records what has been built and is live as of October 2026.

1.1 Purpose

This document describes the functions, users, data and technical setup of the ERP. It is the reference for the owner, office staff, future developers and anyone auditing the system.

1.2 Scope

The system covers:

- Real estate projects, floors, unit categories and units (flats, shops, offices)
- Bookings, installment plans, transfers, cancellations and possession
- Customer receipts, cheques, refunds and printed letters/slips
- Rentals, land and plot trading
- Material inventory, purchase orders, GRNs, warehouses and site dispatch
- Contractors, agreements, bills and payments
- Office expenses, wages, owner personal expenses and petty cash
- Partners, their capital, drawings and project shares
- Chart of accounts, vouchers and financial reports
- Leads, SMS/WhatsApp communication, brokers and commission
- A customer portal where buyers see their bookings and receipts

Out of scope: payroll tax, bank reconciliation feeds and online payment collection.

1.3 Audience

Reader	Uses this document to
Owner / management	Confirm what the system does and sign off

Reader	Uses this document to
Office staff	Understand each module before training
Developers	Maintain and extend the code
Auditors	Trace how money and stock are recorded

1.4 Definitions

Term	Meaning
Allottee	The person a unit is allotted to (the buyer of record)
Booking	A sale of one unit to a customer, with its payment plan
Installment plan	The schedule of amounts a buyer owes and their due dates
GRN	Goods Received Note, records material arriving from a vendor
PO	Purchase Order sent to a vendor
Petty cash float	A cash amount held by a staff member for small spends
Voucher	An accounting entry: Receipt, Payment, Journal or Contra
Broker / booking agent	A person who brings a sale and earns commission

2. Overall description

The ERP is a browser app for office staff plus a separate portal for customers, backed by one MySQL database.

2.1 User classes

User	How they sign in	What they can do
Admin	Staff login (username + password)	Everything, including users, roles, settings and backups

User	How they sign in	What they can do
Staff	Staff login	Only the modules their role is given
Customer	Customer portal login (sign up, forgot password)	View own bookings, receipts and refunds; print receipts; change password

Roles are created by the admin. Each role is a list of module keys; a staff user sees only those modules in the menu and the API refuses the rest.

2.2 Technology

Layer	Technology
Frontend	React 19, TypeScript, Vite, Tailwind CSS 4, React Query, Zustand, React Router 7
Backend	Python, FastAPI 0.115, SQLAlchemy 2, Pydantic 2, Alembic migrations
Database	MySQL 8
Auth	JWT tokens (python-jose), bcrypt password hashing
Messaging	Twilio for SMS and WhatsApp
Other	QR codes on printed documents, Excel import/export (openpyxl), image upload

2.3 Operating environment

- Runs in any modern browser on desktop and mobile; printed documents use the browser's print / Save as PDF.
- Hosted on Railway: one frontend service, one backend service, one MySQL service.
- Staff session lasts 8 hours (480 minutes) before login is needed again.

2.4 Constraints and assumptions

- Currency is Pakistani Rupees (PKR); areas use Sq. Ft., Sq. Yd., Marla and Kanal.
- One company per installation; company name, logo and details come from Company Settings.
- Database schema changes go only through Alembic migrations, run automatically on each backend deploy.

3. Functional requirements

The ERP's modules fall into 13 groups. Each requirement below is built and live; IDs (FR-x.y) are for reference in change requests.

3.1 Projects and units

- FR-1.1 Create, edit and delete projects; group projects under project groups; mark a project Active or Inactive.
- FR-1.2 Define floors per project and unit categories (e.g. Shop, Flat, Office).
- FR-1.3 Create units with project, floor, category, size and price; unit status is Available, Booked, Sold, On-Hold, Rented or Cancelled.
- FR-1.4 Set a project payment template (down payment + installment lines) used as the default plan for new bookings.
- FR-1.5 Record partner shares per project.
- FR-1.6 Print unit lists; view project detail with its units and deliveries.

3.2 Customers, allottees and bookings

- FR-2.1 Register allottees (buyers) with contact and identity details.
- FR-2.2 Book a unit for an allottee: price, down payment, installment frequency (Monthly, Quarterly, Half-Yearly, Yearly) and the generated payment schedule.
- FR-2.3 Add extra charges to a booking (East/West/Road facing, Corner, Open, Water, Electricity, Documents, Other).
- FR-2.4 Move a booking through Booked → Confirmed → Possession Given, or Cancelled.
- FR-2.5 Transfer a booking to a new allottee and keep the transfer history.
- FR-2.6 Assign a broker/booking agent to a booking.
- FR-2.7 Print booking letter and allotment letter.

3.3 Receipts and refunds

- FR-3.1 Record a receipt against a booking for Booking, Installment, Extra Charges or Documentation Charges; one receipt can be split across several installments.
- FR-3.2 Payment modes: Cash, Cheque, Bank Transfer, Online.
- FR-3.3 Track cheques as Pending, Cleared or Bounced, with a pending-cheques list.
- FR-3.4 Print receipts and payment slips with QR code.
- FR-3.5 Create refunds for a Customer, Vendor or Employee; apply deductions; pay in parts (Pending → Partially Paid → Paid); print refund slips.

3.4 Rentals

- FR-4.1 Register tenants and rent agreements on units or land; agreement status Active, Terminated or Expired.
- FR-4.2 Generate the rent schedule and record rent receipts against it.
- FR-4.3 Rent on a project's unit or shop counts as that project's income.
- FR-4.4 Print rent agreements and rent receipts.

3.5 Land and plots

- FR-5.1 Record land properties (Plot, Land, Commercial Shop, SR, Other) with size in Sq. Yd., Sq. Ft., Marla or Kanal.
- FR-5.2 Track status Available, Reserved, Sold, Rented.
- FR-5.3 Record payments To Seller (purchase) and From Buyer (sale).

3.6 Material inventory and procurement

- FR-6.1 Manage vendors, warehouses and materials.
- FR-6.2 Enter opening stock per warehouse.
- FR-6.3 Raise purchase orders (Draft → Approved → Closed, or Cancelled).
- FR-6.4 Receive goods through GRNs, which add stock.
- FR-6.5 Issue material to a project site (Site Consumption or Damaged/Wastage); issue status Dispatched → Received at site.
- FR-6.6 Transfer stock between warehouses.
- FR-6.7 Every movement writes a stock ledger line (GRN, Issue, Restock, Transfer, Opening, Petty Cash); view stock by warehouse and by project.
- FR-6.8 Vendor refunds pick the vendor and GRN and show as Received.

3.7 Contractors

- FR-7.1 Register contractors.
- FR-7.2 Create contract agreements per project on Lump Sum, Per Sq. Ft., Per Sq. Yd., Per Floor or Item Rate/BOQ basis; status Active, Completed, Cancelled.
- FR-7.3 Record contractor bills and payments (Advance, Bill Payment, Retention Release); print payment slips.

3.8 Expenses and petty cash

- FR-8.1 Office expenses, with print.
- FR-8.2 Employees and wage payments (Monthly or Daily wage).

- FR-8.3 Owner's personal expenses, kept apart from company expenses.
- FR-8.4 Petty cash floats per holder, top-ups and spends; spends made on other screens show in petty cash with the holder's name.

3.9 Partners

- FR-9.1 Register partners and their share in each project.
- FR-9.2 Record partner contributions (capital in), drawings (money out) and expenses paid by partners.
- FR-9.3 Partner summary and printable partner statement.

3.10 Accounts and vouchers

- FR-10.1 Chart of accounts with Asset, Liability, Capital, Revenue and Expense heads; a default chart is seeded on install.
- FR-10.2 Vouchers of type Receipt, Payment, Journal and Contra, with printable voucher.
- FR-10.3 Operational screens (receipts, expenses, stock, partner entries) post to the accounts so reports stay in step.

3.11 Leads, communication and brokers

- FR-11.1 Capture leads by hand or import from a file.
- FR-11.2 Send SMS or WhatsApp messages to one person or in bulk, linked to a receipt, booking or refund; each message is logged as Sent or Failed.
- FR-11.3 Manage brokers (booking agents), view their bookings summary, and pay commission through commission payouts with print.

3.12 Customer portal

- FR-12.1 Customer sign-up, login, forgot password and change password.
- FR-12.2 Customer dashboard showing their bookings and balances.
- FR-12.3 List and print their own receipts; view their refunds.
- FR-12.4 A customer sees only their own records.

3.13 Users, roles and administration

- FR-13.1 Admin creates users and roles and picks which modules each role can open.
- FR-13.2 Company settings (name, address, logo, contact) used on every printed document.
- FR-13.3 Admin can download a database backup.

- FR-13.4 Global search across records.
- FR-13.5 Upload images (e.g. logo, documents).

4. Reports and dashboard

The ERP produces 15 reports, each viewable on screen and printable; the dashboard gives the day's position at a glance.

Area	Report	Shows
Accounts	Trial Balance	Debit and credit totals of every account
Accounts	Profit & Loss	Revenue minus expenses for a period
Accounts	Balance Sheet	Assets, liabilities and capital on a date
Accounts	General Ledger	Every entry in one account, with running balance
Sales	Aging	Customer dues by how overdue they are
Sales	Customer-wise	Each customer's bookings, paid and balance
Sales	Sales & Purchase	Sales against purchases for a period
Sales	Brokers	Bookings and commission per broker
Stock	Stock	Current stock per material and warehouse
Stock	Stock Ledger	Every movement of one material
Stock	Materials	Material list and usage
People	Partners	Each partner's contributions, drawings and share
People	Employees	Wages paid per employee
Property	Rental Income	Rent received per agreement and project
Property	Land Properties	Land bought, sold and payments made/received

Printed outputs besides reports: receipt, payment slip, booking letter, allotment letter, refund, rent agreement, rent receipt, voucher, expense, petty cash, contractor payment, commission payout, partner statement, units list, inventory and invoice.

5. Data model overview

The database holds 66 tables; the project is the hub that units, bookings, stock, contractors and partner shares hang from.

Area	Main tables	Key links
Projects	projects, project_groups, project_floors, unit_categories, units, project_payment_templates, project_payment_template_lines	unit → project, floor, category
Sales	allottees, customer_accounts, bookings, booking_installment_plans, payment_schedule_lines, booking_extra_charges, booking_transfers	booking → unit, allottee, broker
Collections	receipts, receipt_allocations, refunds, refund_payments	receipt → booking; allocation → schedule line
Rentals	tenants, rent_agreements, rent_schedule_lines, rent_receipts, rent_receipt_allocations	agreement → tenant, unit or land
Land	land_properties, land_property_payments	payment → property
Inventory	vendors, warehouses, materials, purchase_orders, purchase_order_lines, grns, grn_lines, material_issues, material_issue_lines, material_transfers, opening_stocks, stock_ledger	ledger line → material, warehouse, source document
Contractors	contractors, contract_agreements, contractor_bills, contractor_payments	agreement → contractor, project
Expenses	office_expenses, employees, wage_payments, owner_personal_expenses, petty_cash_floats, petty_cash_topups, petty_cash_expenses	wage → employee; spend → float
Partners	partners, project_partner_shares, partner_contributions, partner_drawings, partner_expenses	share → partner, project
Accounts	accounts, vouchers, voucher_lines	voucher line → account
CRM	leads, communication_logs, booking_agents, commission_payouts	payout → broker; log → user

Area	Main tables	Key links
System	users, roles, role_module_permissions, company_settings, sequence_counters, alembic_version	user → role

sequence_counters issues document numbers (receipts, vouchers, bookings) in order; alembic_version records the schema version.

6. Non-functional requirements

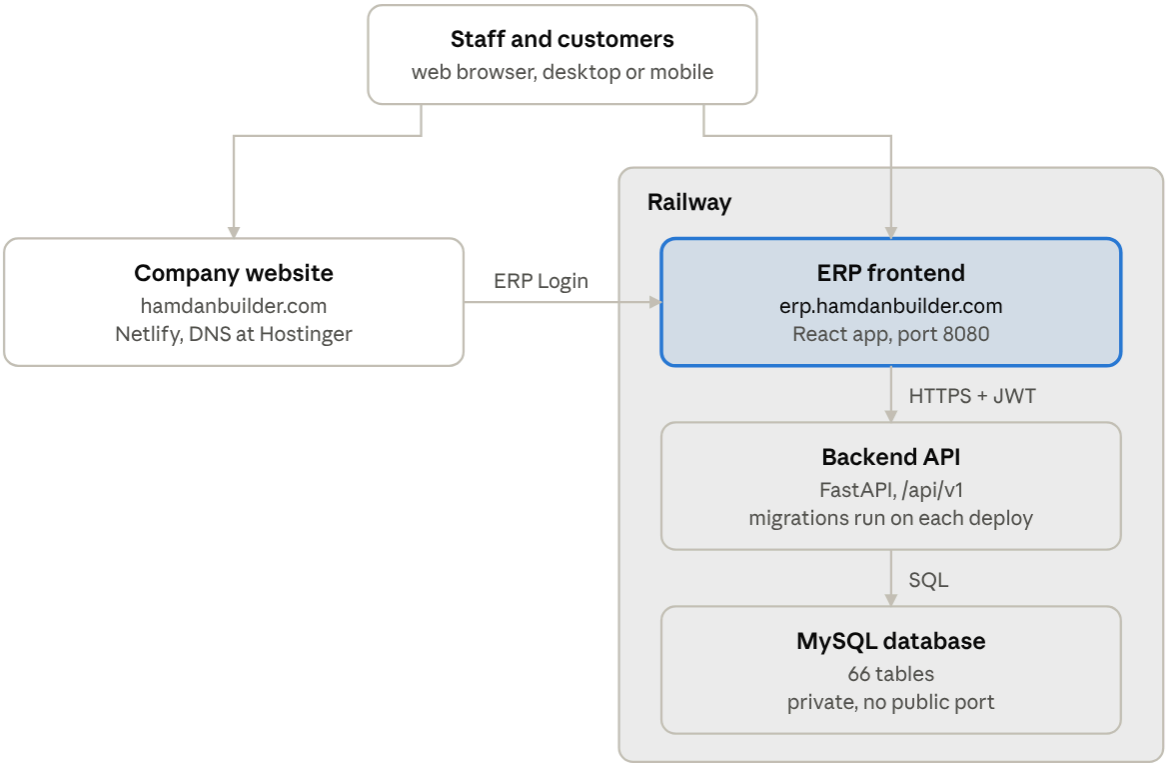
Area	Requirement
Authentication	Staff and customers sign in with password; passwords stored only as bcrypt hashes; access by JWT token that expires after 8 hours
Authorisation	Every API call checks the user's role and module permission; customers can read only their own bookings, receipts and refunds
Separation	Staff and customer accounts are separate; one cannot be used as the other
Cross-site access	The backend accepts browser calls only from listed ERP addresses (CORS)
Data integrity	Foreign keys between tables; schema changes only by migration
Numbering	Receipts, vouchers and other documents get sequential numbers
Backups	Admin can download a backup from the app; Railway's MySQL Backups tab is available; a JSON backup is taken before any bulk data wipe
Secrets	Database passwords and signing keys live in Railway variables, never in the code repository
Testing	Backend has an automated pytest suite covering bookings, refunds, inventory, partners, contractors, payments and more
Usability	Works on desktop and mobile browsers; every printable document fits A4 via the browser's print

Area	Requirement
Availability	Hosted on Railway with automatic redeploy from GitHub; HTTPS on all public addresses

7. Deployment and domains

The ERP is live on Railway at erp.hamdanbuilder.com; the company website's "Customer ERP Login" button opens it.

The ERP runs on Railway; the website links to it



deployment · website, 3 Railway services

Only the frontend and backend are reachable from the internet; the database is reached by the backend inside Railway, or by an admin through a Railway SSH tunnel.

Item	Value
ERP address	https://erp.hamdanbuilder.com (old: elegant-creativity-production-61d2.up.railway.app)
Backend address	hamdanerpanddevelopers-production.up.railway.app/api/v1

Item	Value
DNS for erp	CNAME erp → zfjryyhp.up.railway.app, plus TXT _railway-verify.erp (Hostinger)
Code	GitHub; a push to main redeploys on Railway
Backend start	alembic upgrade head, then uvicorn on Railway's port
Frontend start	Vite build served by serve on port 8080
Allowed origins	Backend CORS lists the old Railway address and erp.hamdanbuilder.com
Data reset	All test data was cleared on 13 Oct 2026; logins, roles, settings and the default chart of accounts were kept